

Metro Ticket Generating System

IDEATION PHASE

Date	25 July 2026
Name	T V SAI LAXMI
Project Name	Metro Ticket generating System
Maximum Marks	2 Marks

1. Problem Statement

1.1 Background

Metro rail is one of the most widely used modes of urban public transport because it is fast, affordable, and reduces road congestion. However, the process of buying a metro ticket in most systems still depends heavily on manual counters, token machines, or partially digital kiosks located inside the station.

Commuters are required to physically visit the station, stand in a queue, choose the source and destination stations, pay in cash or card, and collect a physical token or paper ticket before they are allowed to enter the platform. This dependency on manual, in-person processes creates friction at every step of the commuter's journey and limits the scalability of metro operations during high-demand periods.

1.2 Problem Statement

“Commuters travelling by metro rail face long queues, manual fare calculation errors, limited counter timings, and dependency on physical tokens/tickets, which leads to delays, overcrowding at ticket counters, and an inconvenient travel experience — especially during peak hours and for first-time or differently-abled passengers.”

1.3 Problem Statement Canvas (Who / What / Why)

Element	Description
Who (I am)	Daily commuter, student, tourist, senior citizen, or differently-abled passenger using metro services. Daily commuters, students, tourists, senior citizens, and differently-abled passengers using metro services.
Trying to	Book a metro ticket quickly and conveniently for their journey.
But	They encounter long queues, manual ticketing procedures, and limited counter availability.
Because	The existing system relies on physical ticket counters, manual fare calculation, and paper or token-based ticketing instead of a fully digital process.



PROBLEM STATEMENT CANVAS

Summarizing the Commuter's Situation



ELEMENT	DESCRIPTION	CURRENT SITUATION (PAIN)	IMPACT
 WHO (I am)	A daily office commuter / student / occasional traveller / senior citizen / differently-abled passenger using the metro.	- I have to physically go to the station to buy a ticket. - I face long queues at ticket counters. - I struggle with vending machines which may be slow, offline or hard to use.	 Wastes time and causes inconvenience.
 WHAT (Trying to)	Book a metro ticket quickly for a specific source and destination station without standing in a queue.	- The process involves multiple manual steps inside the station. - Fare calculation is done manually. - Payment and ticket issuance are slow.	 Leads to delays and affects productivity.
 BUT (But)	I am forced to wait in long queues at ticket counters or struggle with vending machines that are slow, offline or not accessible.	- Counters get crowded during peak hours. - Machines may be out of service. - Process is difficult for elderly, disabled and first-time users.	 Creates frustration and a poor travel experience.
 WHY (Because)	Ticket booking is largely manual, counter staff availability is limited during peak hours, and fare calculation is done manually.	- High dependency on manual processes. - Limited operating hours of counters. - No real-time digital system for quick booking and ticket generation.	 Increases operational costs and reduces overall efficiency.



IN SHORT: Commuters want a fast, simple and accessible way to book metro tickets, but manual processes, long queues and system limitations make the experience slow, frustrating and inefficient for everyone.

Figure 1: Problem Statement Canvas summarizing the commuter's situation

1.4 Key Points

- Long waiting time at ticket counters.
- Manual fare calculation increases the possibility of errors.
- Physical ticket loss causes inconvenience.
- Limited operating hours of ticket counters.
- Poor accessibility for elderly and differently-abled passengers.
- High operational and maintenance costs.
- Lack of digital travel records.
- Slow refund and complaint resolution.

1.5 Impact Across Stakeholders

Passengers

- Long waiting time
- Travel delays
- Poor user experience
- Difficulty for elderly users

Metro Authorities

- High operational cost

- Increased staffing requirements
- Difficult crowd management
- Manual reporting

Employees

- Heavy workload
- Repetitive tasks
- Higher chances of errors

Environment

- Paper waste
- Plastic token usage
- Increased maintenance

1.6 Need for a Digital Solution

There is a clear need for a digital, self-service metro ticket booking system that lets commuters select their stations, get an automatically calculated fare, and instantly receive a QR-code based digital ticket — all without visiting a counter. Building this on the ServiceNow platform allows the solution to leverage the Service Catalog for a guided booking experience, Flow Designer for automated fare calculation and ticket generation, and Notifications for instant delivery of the digital ticket, resulting in a faster, more accessible, and more reliable travel experience.